



UNIVERSITY OF AMSTERDAM

Recommendation Systems

Mobile Marketing: Mobile-Apps & Geo-Targeting

Digital Marketing and Analytics: Week 6

Academic Year 2025-2026 – Semester 1 – Block 2

AMSTERDAM
BUSINESS
SCHOOL

Business

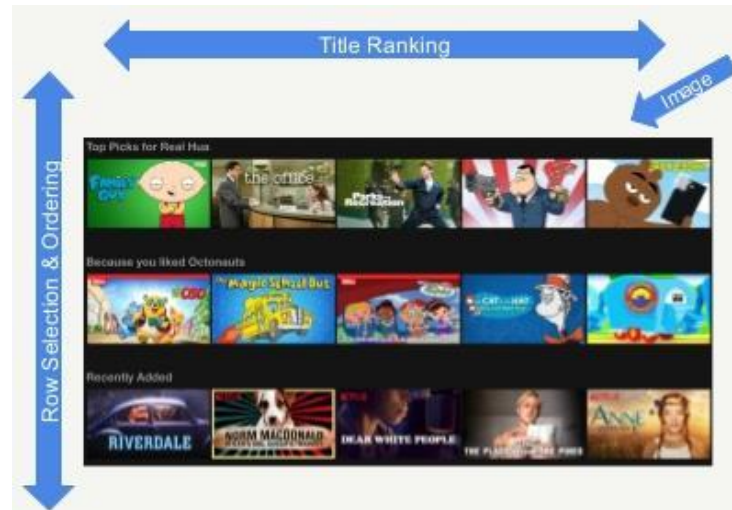
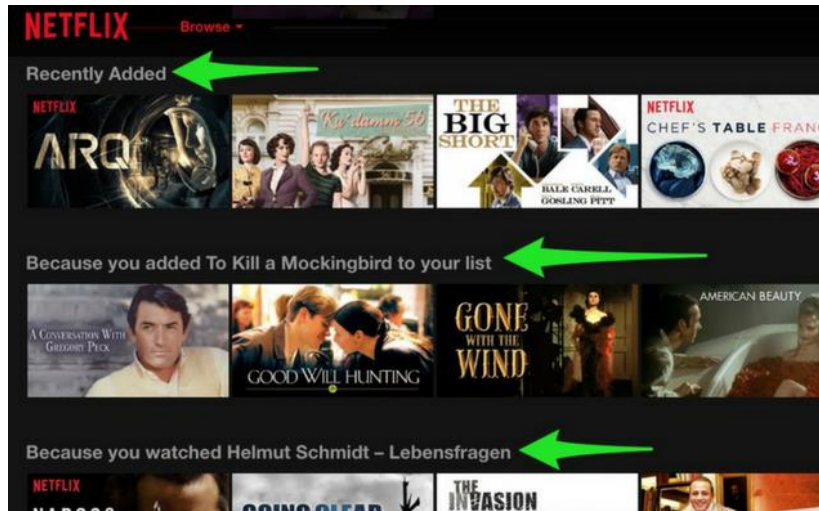
Personalization: Recommender Systems

Personalization on Platform: Recommendation Systems

Personalized advertising is useful to attract-create traffic and new members to your website-platform

But what to do when the customer is once/already on your website-platform as a user? Then the work is taken over by RECOMMENDER systems.

Personalized recommendations account for > 75% of all YouTube and Netflix traffic and > 70% of new song discoveries on Spotify.



Personalization with Recommendation Systems

Recommendation Engines (RE) runs on machine learning algorithms

Recommender Engines **enhance customer engagement, increase average order value, improve up/cross-selling opportunities**

There are different forms and make recommendations on various algorithms (decision rules)

改善追加/交叉销售机会

Customers who viewed this item also viewed these products

			
Dualit Food XL1500 Processor \$560 Add to cart	Kenwood kMix Manual Espresso Machine ★★★★★ \$250 Select options	Weber One Touch Gold Premium Charcoal Grill-57cm \$225 Add to cart	NoMU Salt Pepper and Spice Grinders \$3 View options

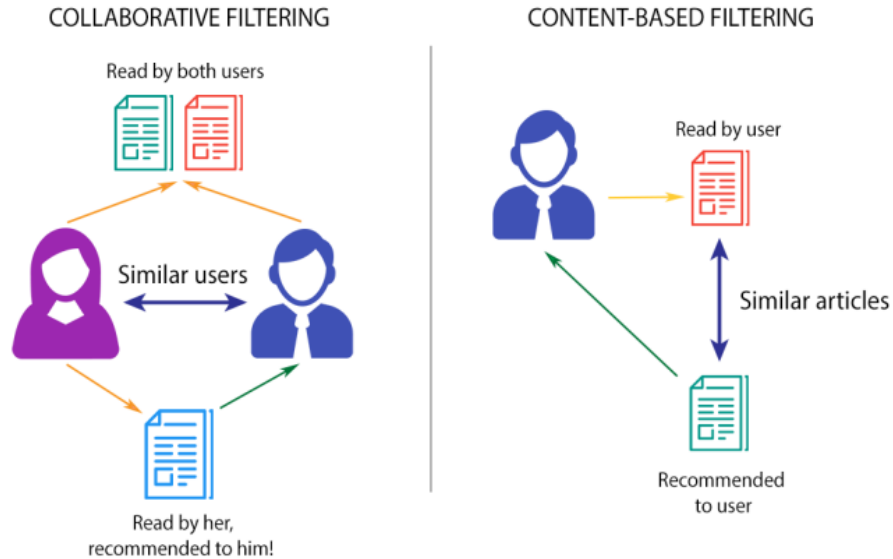
People you may know

 Alice Wilson Employer Engagement Executive for... 26 mutual connections Connect	 James Logie Global Retail Specialist: International... 5 mutual connections Connect	 Julia Roberts Social Media Manager & Marketing Trainer 24 mutual connections Connect	 Ria Goff Finding alternate routes of finance for busines... 3 mutual connections Connect
 David Leversha Buildings - Structural Director at SWECO... 4 mutual connections Connect	 Carolann Creane Senior Commercial Recruitment Consulta... 2 mutual connections Connect	 Imre Tamas Toth CEO at large Corporate Growth an... 4 mutual connections Connect	 Caroline Skepper Interior Designer at Caroline Skepper... Based on your profile Connect

Recommendation Systems: Main Types

Although there are multiple types, recommendation systems (filters) are usually categorized in two main kinds:

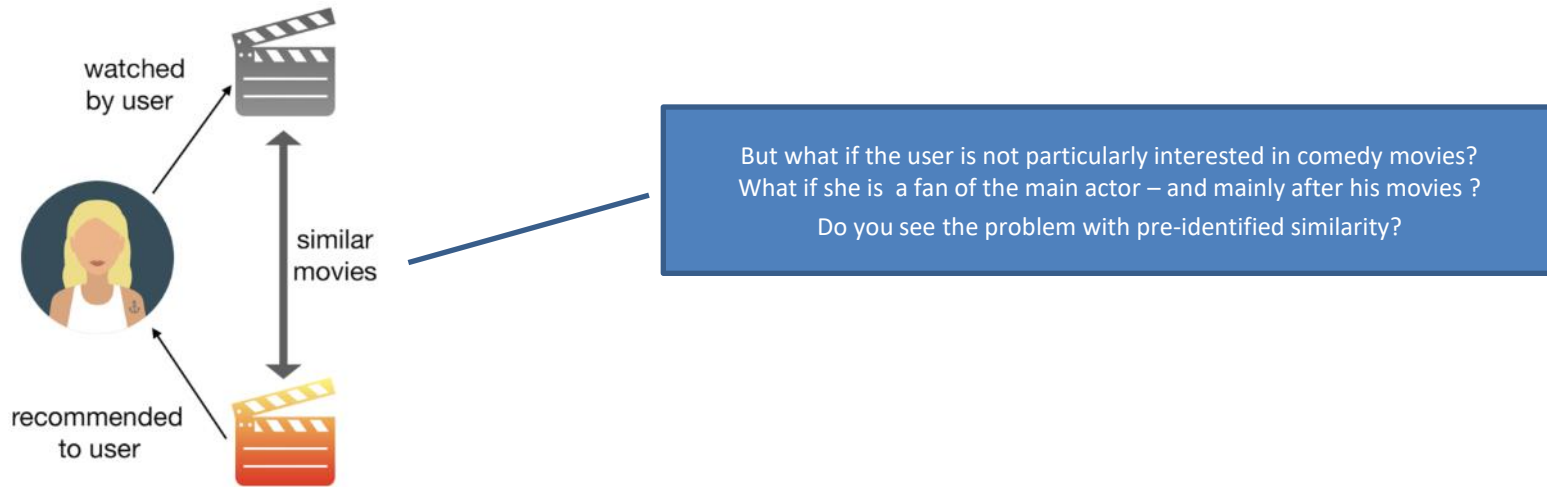
(1) Collaborative <user based> filtering models and (2) Content <item-product based> based models.



Recommendation Systems: 1 Content (Item) Based Filtering

“You bought this (X) you may also be interested in that (Y) – that is similar”

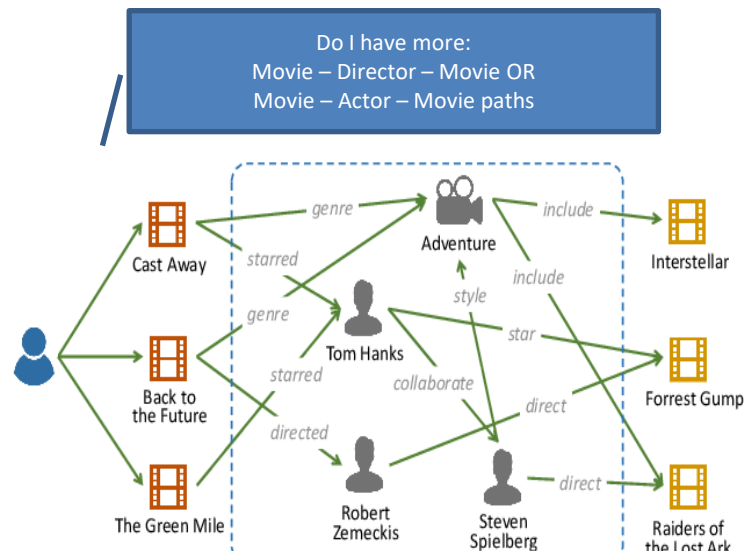
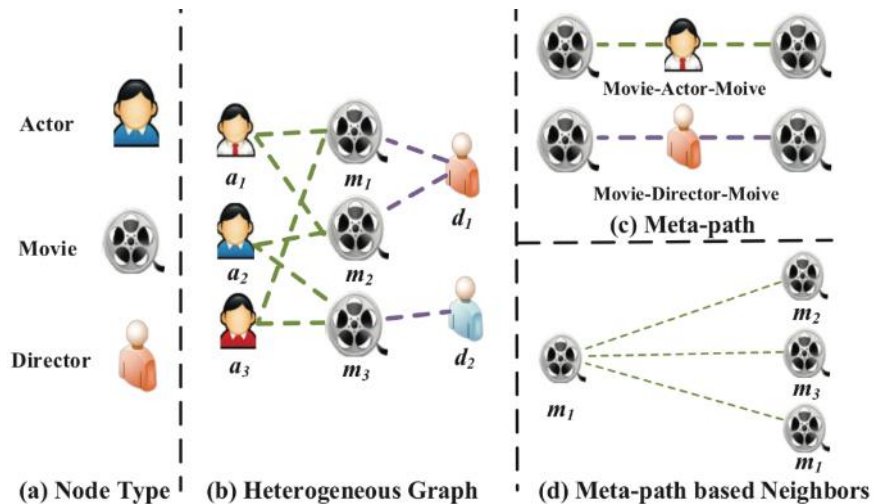
- Users who like a product will probably like a similar product.
- Content Based filter recommends a new product similar to purchased product
- Similarities are pre-identified by the firm and/or analysts (comedy movie is considered as similar with another comedy movie)



Recommendation Systems: 1 Content (Item) Based Filtering

To improve the item (content) based recommendations and link with user based matching – one can use **meta-paths** (see below):

A **Meta Path** is a sequence of relations between object types, which defines a new composite relation



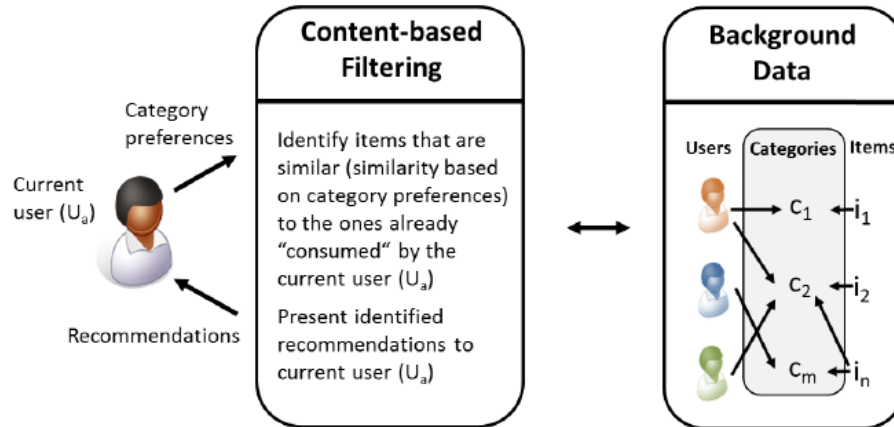
Recommendation Systems: 1 Content (Item) Based Filtering

(1) Pre-defined set of product characteristics: e.g. genre, writer, actors

(2) Calculate similarity between products > (3) Select the products that are most similar to the consumed products

PROS: Simple and intuitive. Useful even if we know nothing about the user (**at cold start**) and if we have little/no data on past purchases.

CONS: Not model based (is high product similarity a good recommendation?) Product characteristics hard to define – what is a similar product?

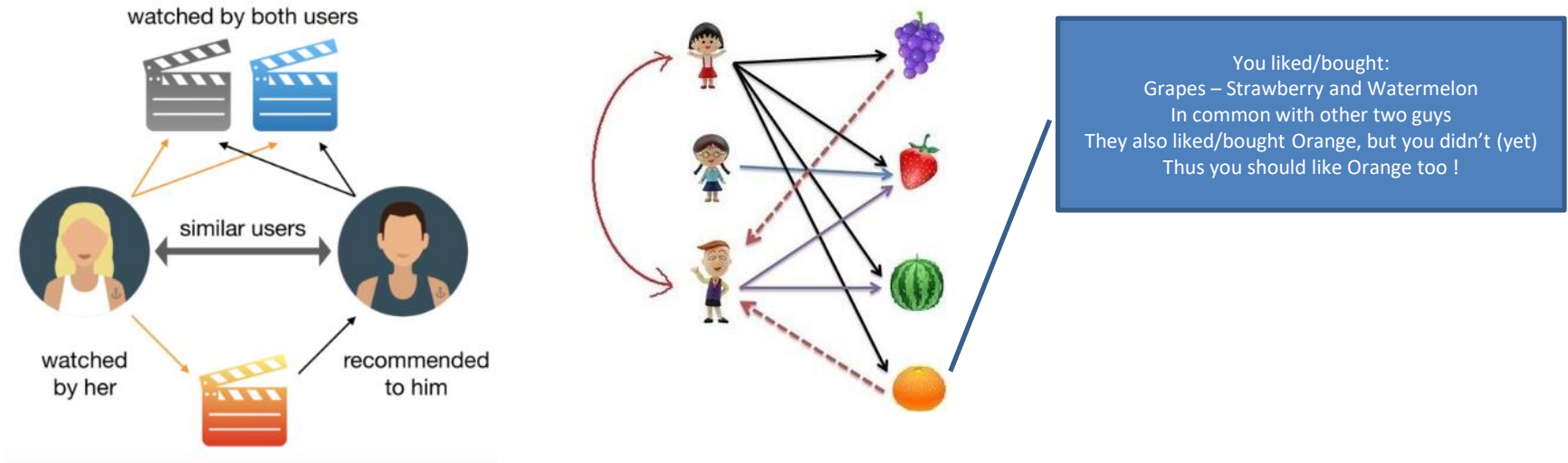


Recommendation Systems: 2 Collaborative (User Based) Filtering

“People LIKE you also bought (saw) this....”

Users will probably like a product if it is liked by a similar user. Recommend a product that similar users like

In other words the recommender will match you with people that **you have a similar liking/ or purchase history** with.



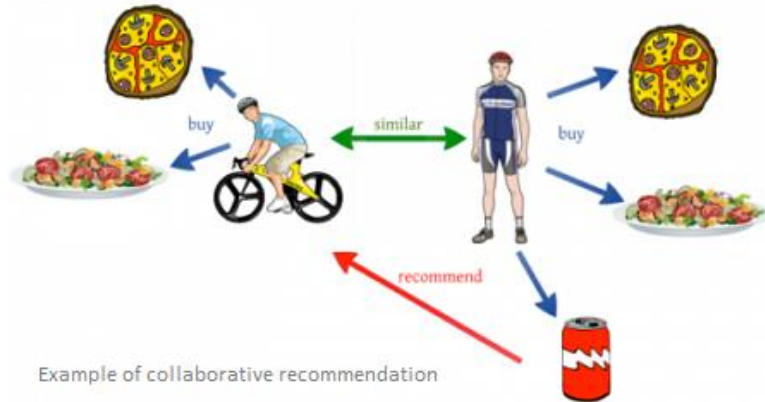
Recommendation Systems: 2 Collaborative (User Based) Filtering

Similarity between the current user and all other users (based on rating behavior) > Select the k users most similar to the current user

PROS: (1) Simple and intuitive (2) Takes into account user behavior

CONS: (1) Not useful for users and products with few data (sparsity)

(2) Not useful for new users and products (cold start) (3) Computation time is large with thousands of users and products

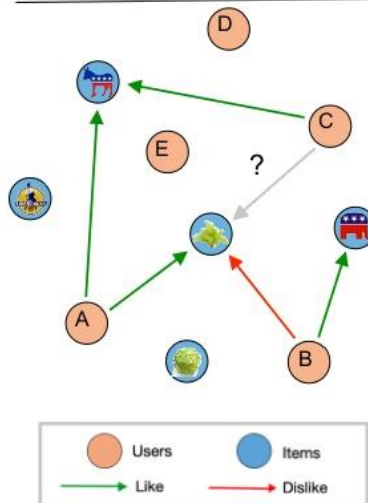


Recommendation Systems: Hybrid or Other Methods

Inferring through Indirect (Paths) Correlations



Inferring correlations



- 1) User A:
 - Democrat
 - Likes *Arugula*
- 2) User B:
 - Republican
 - Dislikes *Arugula*
- 3) User C indicates:
 - Democrat

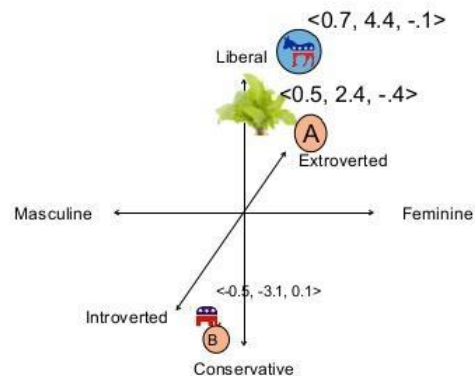
What would we infer is User C's affinity for *Arugula*?

Answer: User C would like *Arugula*

4



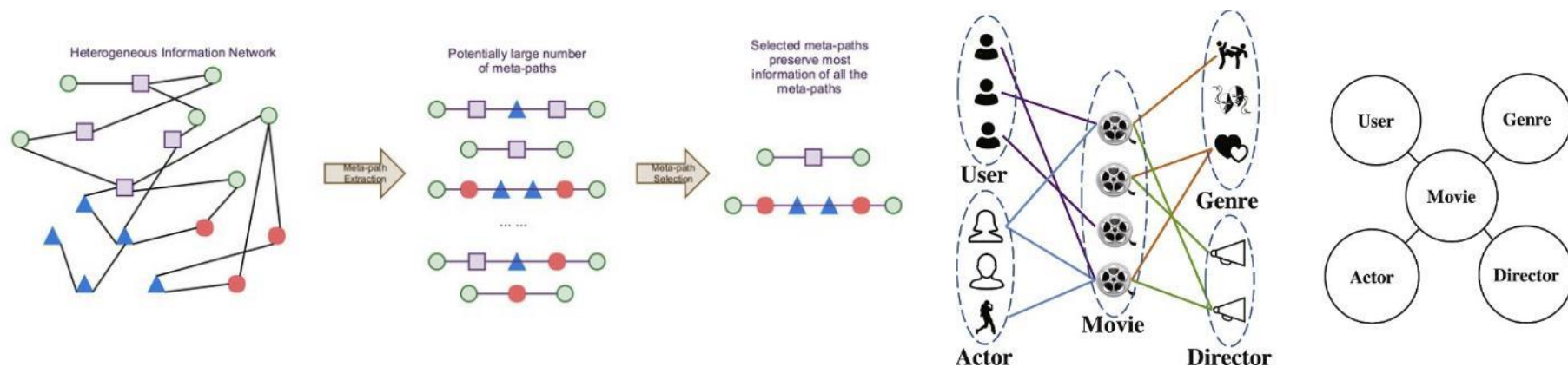
Taste space = many latent factors



Recommendation Systems: Hybrid Logic > Metapaths

Semantic Path Based Learning & Meta-Paths

Meta-Path based approaches combine **user feedback data** with **additional information**, such as item or user attributes and relationships. Therefore, they can emulate collaborative filtering, content-based filtering,



Wei et al (2018) Unsupervised Meta-path Reduction on Heterogeneous Information Networks

Liu, H., Jiang, Z., Song, Y., Zhang, T., & Wu, Z. (2019). User preference modeling based on meta paths. Knowledge-Based Systems, 181, 104784.

Recommender Systems: Framing (Same) Recommendation Differently

Two common framings might describe the same recommendation.

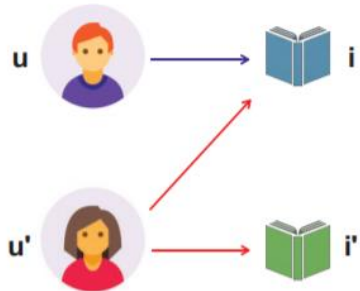
User-based framing emphasizes the similarity between customers (e.g., “**People who like this also like...**”);

Item-based framing instead emphasizes similarities between products (e.g., “**Similar to this item**”).

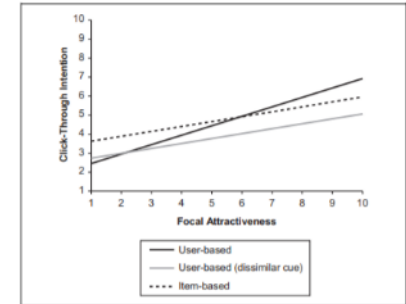
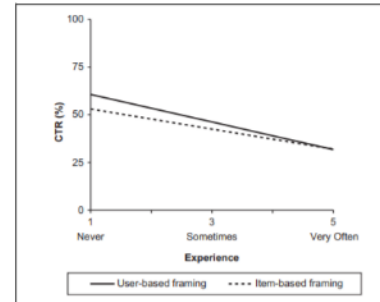
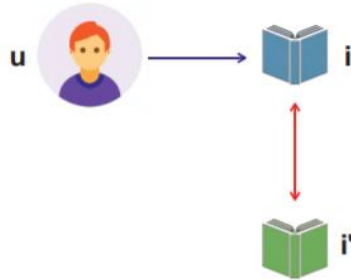
Framing the same recommendation as **USER-BASED (vs. item-based)** can increase recommendation click-through rates. Especially when:

(1) The customer is less experienced (2) attractiveness of the product is higher (3) and the user perceive the similarity with others as high

User-Based Framing



Item-Based Framing



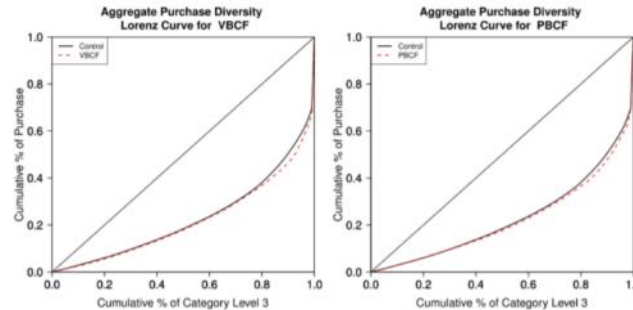
Recommenders: The Effect on Sales Diversity

A Critical View: Collaborative filters (CFs)—**will decrease sales diversity** since they only **reinforce the popularity of already well-known products**.

Collaborative filters (CFs) is associated with a **decrease in sales diversity** relative to no recommendations

If the recommender is based on **purchase data** (rather than view data) this negative effect is stronger.

*Similar users still end up exploring the same kinds of products, resulting in **concentration bias** at the aggregate level.*



Note. These Lorenz curves show that firm-level aggregate sales diversity is lower for both collaborative filtering algorithms relative to the control.

CONTROL
NO RECOMMENDER
VBCF VIEW BASED RECOMMENDER
PBCF PURCHASE BASED RECOMMENDER

LORENZ CURVE:
INDICATOR OF SALES (PRODUCT)
DIVERSITY
MORE CLOSE TO LINEAR = MORE DIVERSE

Recommenders: Human vs AI Recommender

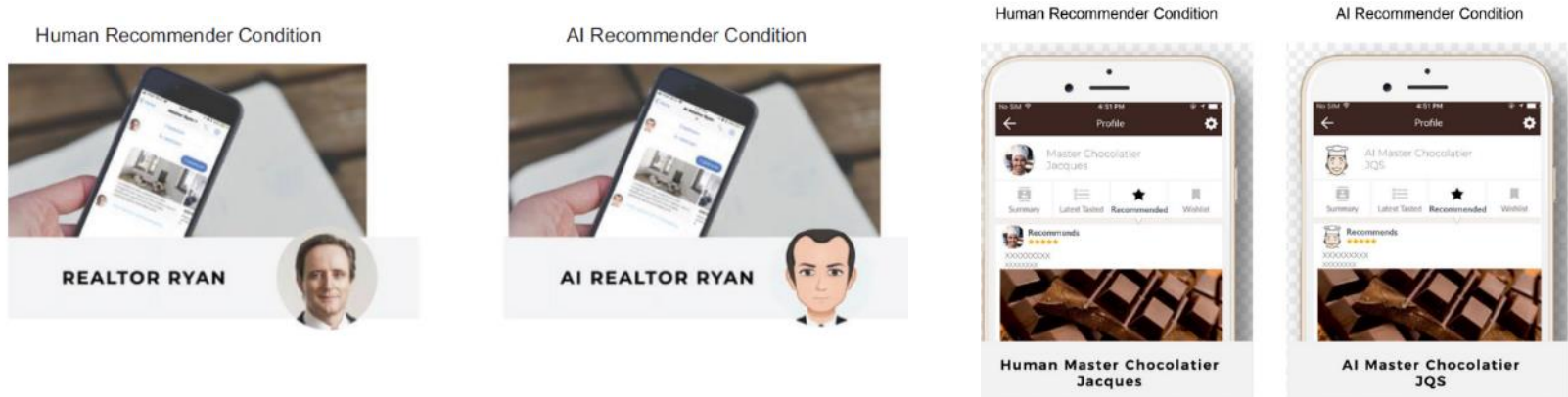
How about AI versus Human (or human-like) recommenders in terms of their impact,

Some of you may think AI recommenders may create a better trust/confidence and may be perceived as algorithmic-more data driven

Some of you may think, customers may feel more “close and warm” therefore may tend to comply with Human recommenders

The answer **can be both** – it depends on whether it is about **Utilitarian or Hedonic** products/realms

- AI recommenders are **more competent** than human recommenders in the **utilitarian realm (utilitarian products)**
- AI recommenders are **less competent** than human recommenders in the **hedonic realm (hedonic products)**



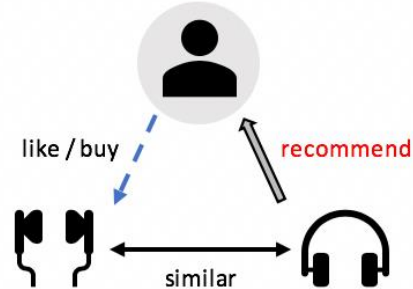
Recommenders: Practice & Self Study Exercise

Self-Study Exercise: Content-Based or Collaborative filtering?

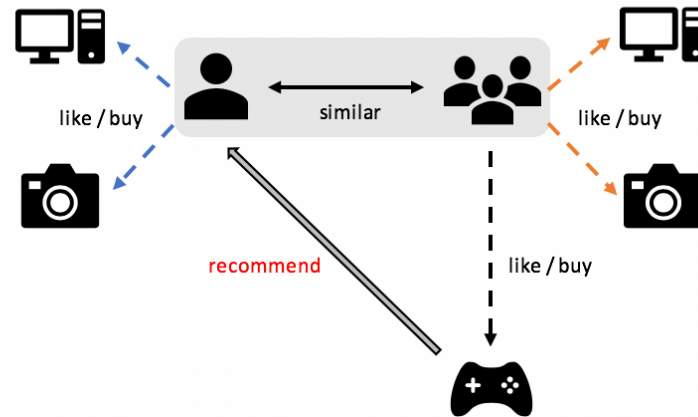
Which one would you prefer for different situations: different products/services?

New products? Services? Durables, movies?

Content Based Recommender System



Collaborative filtering recommender system

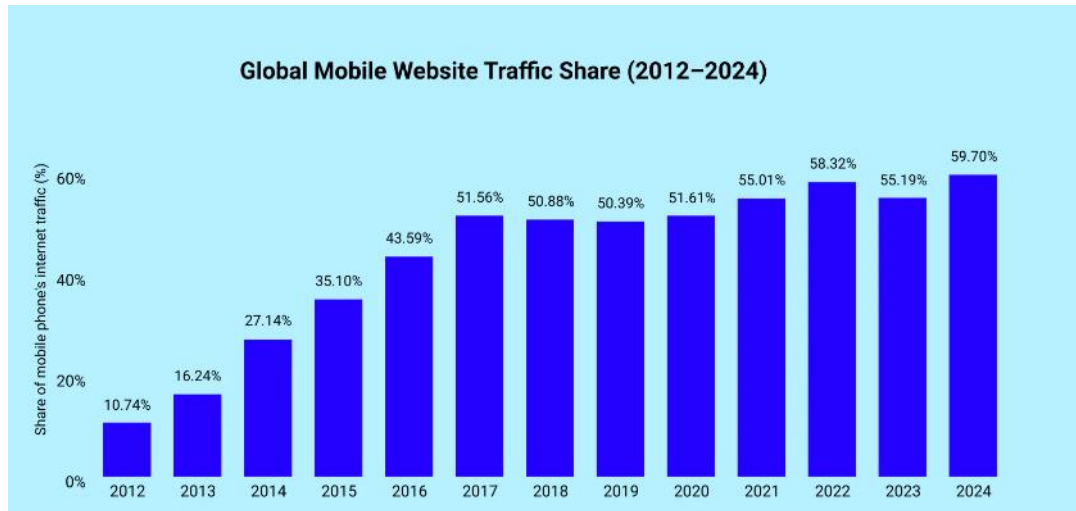


Mobile vs Desktop

Mobile: Dominating Platform

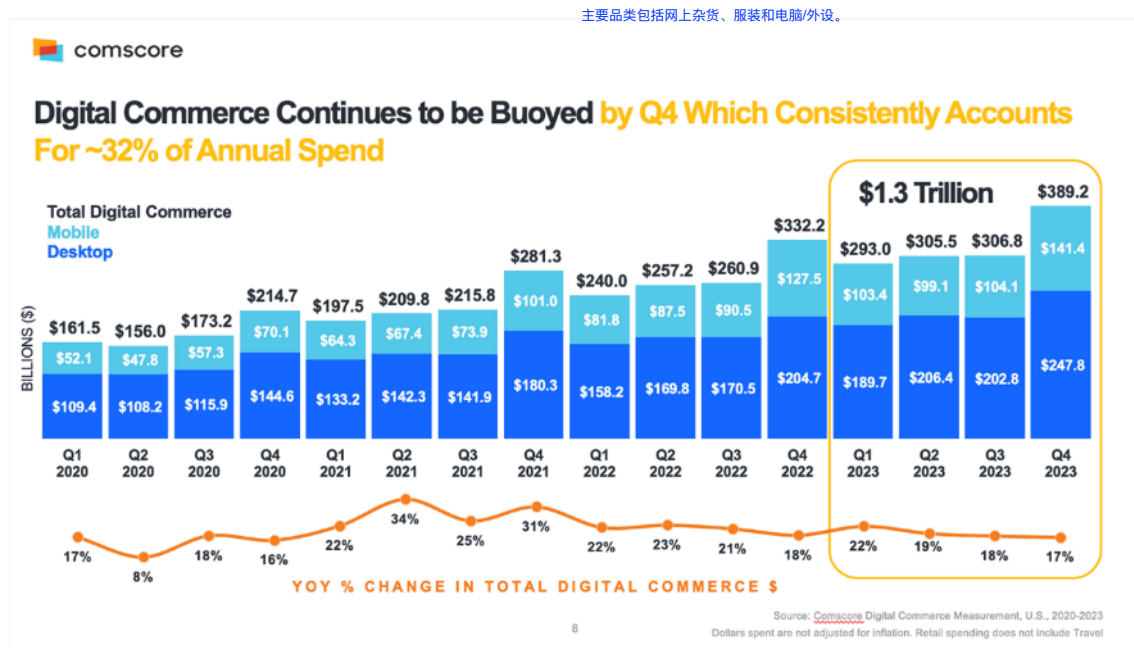
In the last decade, Mobile (devices) has become the new dominating platform in online traffic.

*Mobile vs. Desktop Web Traffic: From 2012 to 2023, global mobile phone website traffic share has seen a significant increase from 10.88% to 60.06%, while desktop share has decreased from 89.12% to 39.94%, **marking a clear shift towards mobile web browsing over the years.***



Mobile vs Desktop: Context matters

Consumer online retail spending hit \$1.3 trillion in 2023, surpassing 2022's first-ever \$1 trillion milestone and growing over 5.5 times in a decade, with leading categories including online grocery, apparel, and computers/peripherals. (Shopper, 2024)



Mobile: What Makes Mobile Different (than Desktop/Laptop)

Mobile Platforms have distinct characteristics that makes them different than non-mobile online:

- 1 App-Based user interface
- 2 On-the-Go nature: Can be used in (physical) stores > bridging offline-online platforms
- 3 Different user characteristics, purposes and motivations / Different drives and consequences before/after usage



How many times per day do you check your mobile phone?

Frequency of Checking Phones: The average internet user checks their phone about 58 times daily, with some Americans checking up to 160 times.

Mobile Has Its Role in Customer Journeys

Mobile Channels and their contributions along customer journeys are important.

However we should NOT overlook the contribution of other (online-offline) channels along customer journeys !

Check-Out: Through which attribution model they are biased below (while interpreting data) – Remember Lecture 1-2 ?

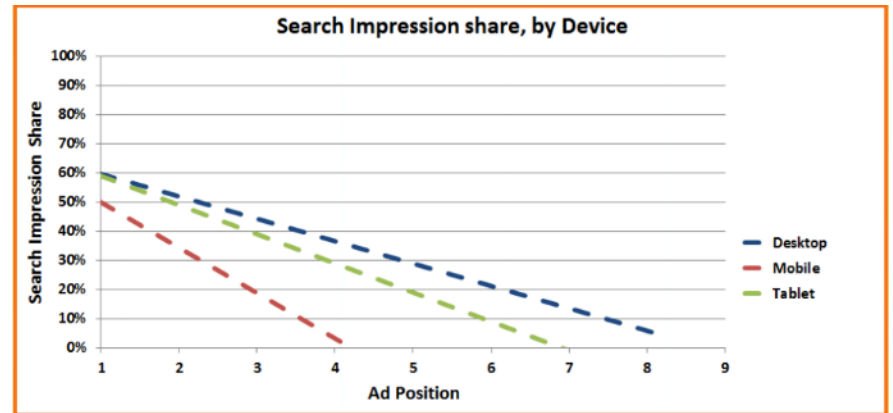
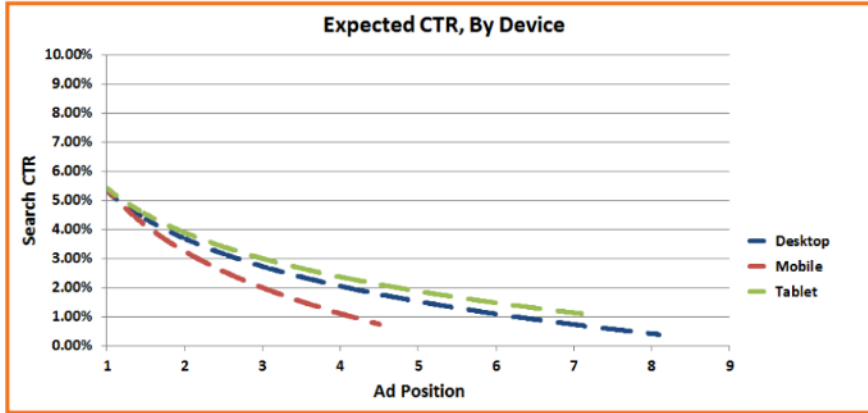


Mobile: Effect of Device-Platform & Time

Mobile: Device Matters – Search (CTR and Impressions)

Search behavior vary depending on the device: non-mobile (desktop) vs mobile (smartphone and tablets)

- Higher CTR for tablets and desktop (> smartphones)
- Highest impression (%) = desktops > tablets > smartphones
- When ad position (# - ranking) goes down: almost no impression on smartphones. Thus it is important to have high ad position # on phones.



Mobile: Device Matters – Conversion Rates

Conversion on mobile devices tracks lower than desktop conversion.

Smartphone conversion low = because people use the device differently (seek information, start a journey etc.)



Mobile vs Desktop: Business Use of Smartphones and Desktops

- Business Video Consumption: Despite the rise of mobile, 87% of business-related videos are viewed on desktops,
- This suggests a preference for larger screens and focused environments in professional settings.
- Desktops for E-commerce Transactions: While mobile devices account for 60% of all e-commerce transactions,
- Desktop visits to e-commerce websites yield a higher conversion rate.

👉 "Mobile 看量, Desktop 看质"

👉 "Mobile browsing, Desktop deciding"



Mobile Vs Desktop

MOBILE DEVICES

Advantages

- Convenience
- Apps
- Multi-function device

Disadvantages

- Screen size
- Mobile functionality on some sites
- Hard to disconnect

DESKTOP DEVICES

Advantages

- Superior performance (in general)
- Keyboards
- Business tools

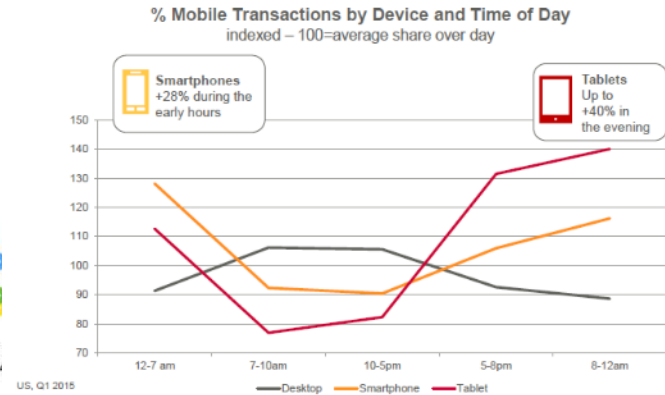
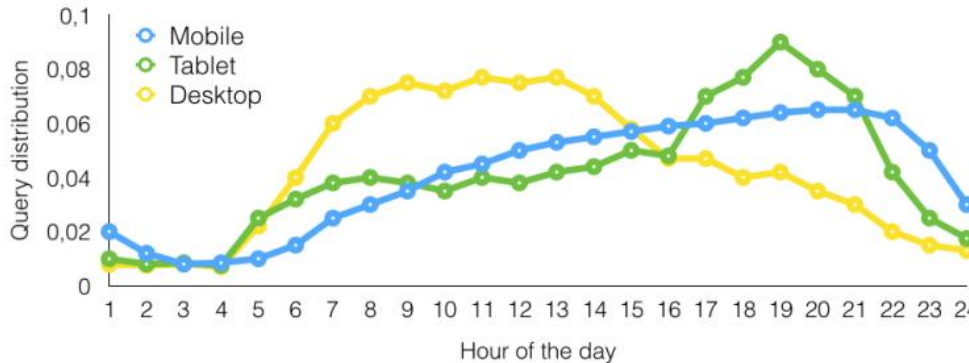
Disadvantages

- Requires more space
- Portability of components
- Outdated over time



Mobile: Time-of-Day Matters Search (per time/device)

- Tablet traffic remained stable throughout the day and shows a peak around dinner time.
- Desktop use/search is associated with day-time (working hours)
- **Tablets are mostly used at home**, and more **specifically in the living room**, followed by the bedroom.



Mobile Apps & Marketing

Drivers of Use and Consequences

Mobile Shopping: To Have or Not to Have an App?

If you Already have a Mobile Website

A KEY QUESTION: For retailers who already offer a **mobile website** is, whether they should add a **mobile APP** as an additional purchase channel?
Whether consumers' app adoption stimulates **additional purchase**?



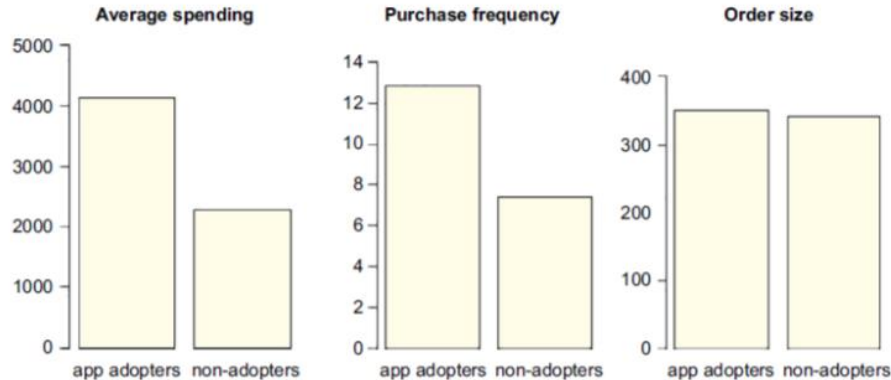
Mobile Shopping: To Have or Not to Have an App?

If you Already have a Mobile Website

Introducing a mobile app is GOOD and PAYS OFF even If when a retailer has mobile website because:

- App adopters have **higher purchase incidence**,
- App adopters buy **more frequently**,
- App adopters **spend more in each order** than non adopters.

These effects are stronger for (a) **low spending customers** (b) **customer who are less loyal to retailer**



Mobile-Apps: Free or Paid – Monetizing a Mobile-Application

Monetization- Commercialization decisions are challenging and could have serious consequence. A developer firm should carefully decide:

Hard-Landing: Pay-or-Churn – there is only a paid version.)

付费或流失

Soft Landing: Has paid option but also a free component. Consumers can pay/get the full service or not pay still get some limited functionality for free.

Besides this should the (app developer) firm offer **Exclusive Secondary Offers** to paying users?

This is a challenging/difficult decision as each choices has advantages and disadvantages when used.



FREE APPS

- ✓ MORE CLIENTS
- ✓ MORE DOWNLOADS
- ✓ MORE VISIBILITY
- ✓ MORE LIKELY POSITIVE REVIEWS
- ✓ LOWER USER EXPECTATIONS

PAID APPS

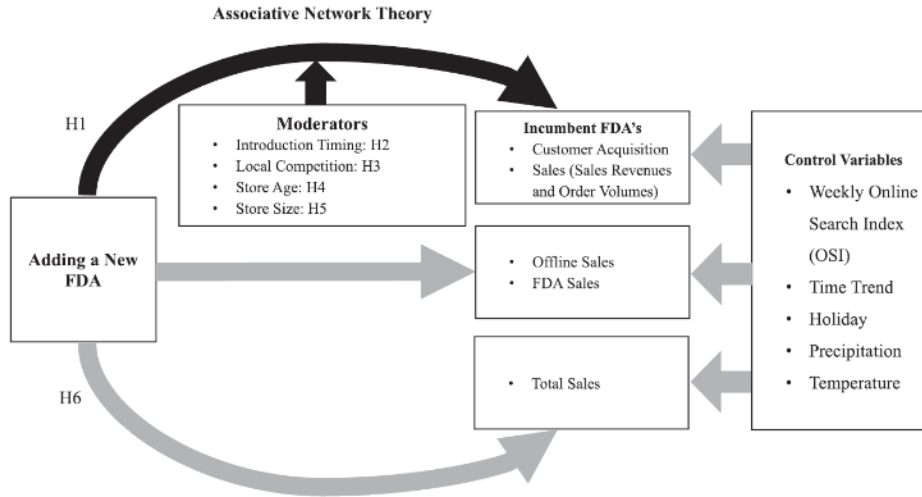
- ✓ LESS COMPETITION ON THE MARKET
- ✓ REVENUE PER DOWNLOAD
- ✓ BETTER LEADS AND ENGAGEMENT
- ✓ LOYAL CUSTOMERS
- ✓ NO ADS

Mobile-Platform Expansion: How Does it Affect Other Channels?

Food Delivery Apps – a popular mobile platform that many restaurants rely on.

When a firm **expands to a new Food Delivery App** to increase its delivery channels:

Would they **cannibalize or enhance** a restaurant's customer acquisition and sales through the existing delivery app. What do you think?



Mobile-Platform Expansion: How Does it Affect Other Channels?

DATA: 557,127 purchase records from 173,537 customers, across 18 store locations of a restaurant chain

Adding a competing **Food Delivery App** has **positive synergistic effects** on restaurant's customer acquisition and sales (revenues and order volumes)

This **effect is stronger** if it adds the competing FDA **long after** (vs. soon after) the prior FDA and the **store is younger** (vs. older).

Larger stores and those located in **competitive areas** benefit more from mobile platform expansion.

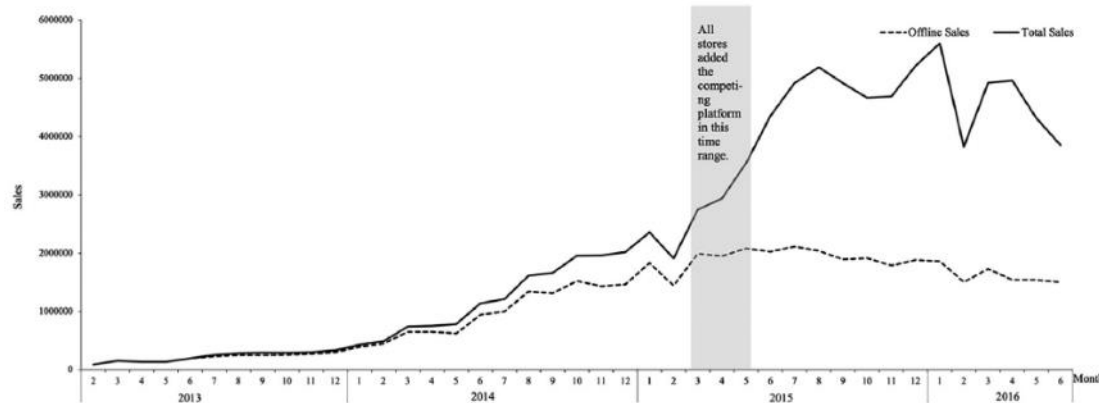


Fig. 4. Offline and total monthly sales at aggregate level.



Mobile-App Failures: Cross-Channel Effects

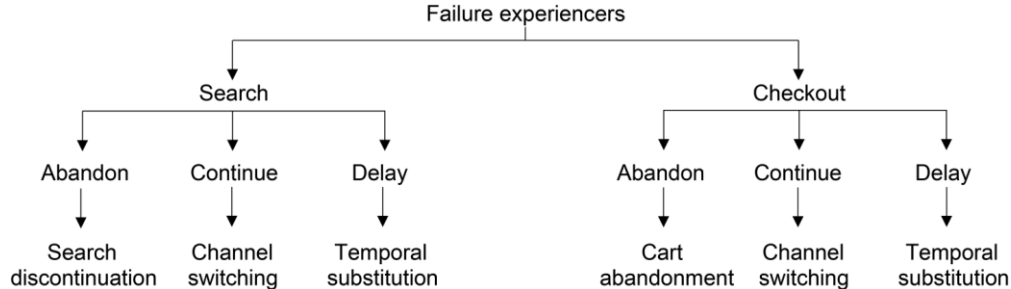
What about a **mobile-app failure** which occurs quite often in business practice.

Researchers run a natural experiment — an **unplanned two-hour outage/failure** of a major retailer's mobile app.

Hypothesis:

App failures will reduce offline purchases (because many users use the app for product search but buy in-store)

App failures will not reduce (and may even shift) online purchases (some users will switch to the website to complete their transaction)



Mobile-App Failures: Cross-Channel Effects

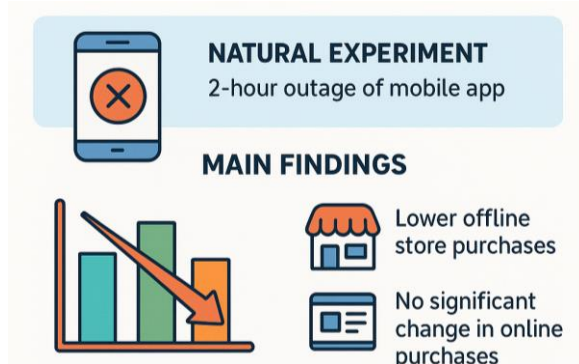
The app failure **significantly decreased overall purchase frequency and quantity.**

This decline was driven by **lower offline store purchases** — the **web/online channel did not show a comparable drop.**

This is due to “search abandonment” (some customers quit purchase if the app fails) and “channel switching” (some shift from app → website).

The effect was **smaller for shoppers who historically spent more with the retailer.**

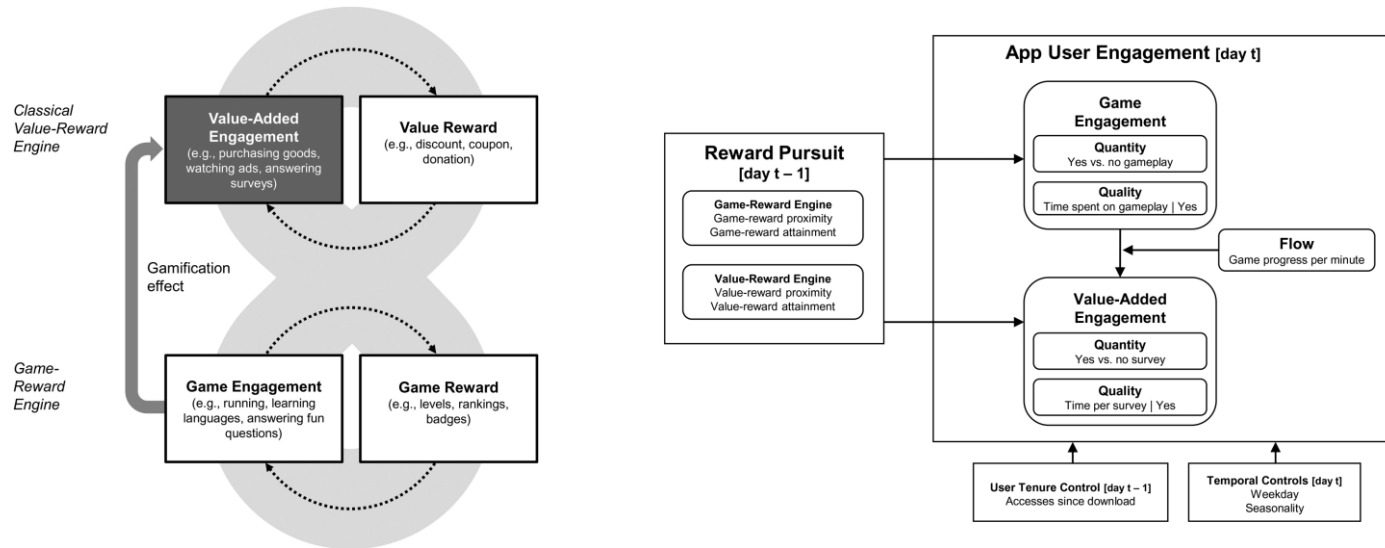
The effect was larger for those who **had more recently engaged (e.g. used app search) with the retailer.**



Customer Engagement through Gamification (Apps)

How adding game-elements (e.g. points, levels, badges) superimposed on traditional “value rewards” (e.g. discounts, coupons) **affects user engagement** in a hybrid reward architecture

Using granular daily data from 18,952 users of a gamified market-research app over a year



Customer Engagement through Gamification (Apps)

Game rewards boost engagement more in comparison to value rewards alone achieve.

The **closer a user is to earning a reward** (in either the game or value), the more their engagement increases (goal-gradient effect).

But there is a “**dark side**”: **when users enter a flow state** with the game (i.e. deeply immersed, highly absorbed), **the translation from game engagement to value-added engagement weakens** (i.e. they focus on the game side, possibly at the expense of value-relevant tasks).

Apps should “gate” (lock) certain **game rewards** until users complete real, **valuable actions first**.

If users only play the daily game but never buy anything, their engagement doesn't help the retailer. So the app could require something like: “You can only unlock the next game level after you've completed a purchase or rated a product.”

如果用户只是玩每日游戏却从不购买任何东西，他们的参与对零售商来说毫无帮助。因此，应用程序可以设置类似这样的提示：“只有完成购买或评价产品后，才能解锁下一关游戏。”

KEY EFFECTS / RELATIONSHIPS



Game rewards
boost
engagement



Closer to reward
→ more engagement



Dark side:
flow state
weakens value-
added engagement

MANAGERIAL / DESIGN IMPLICATIONS



Gamification is
cost-efficient



Avoid
simultaneous
reward
attainment



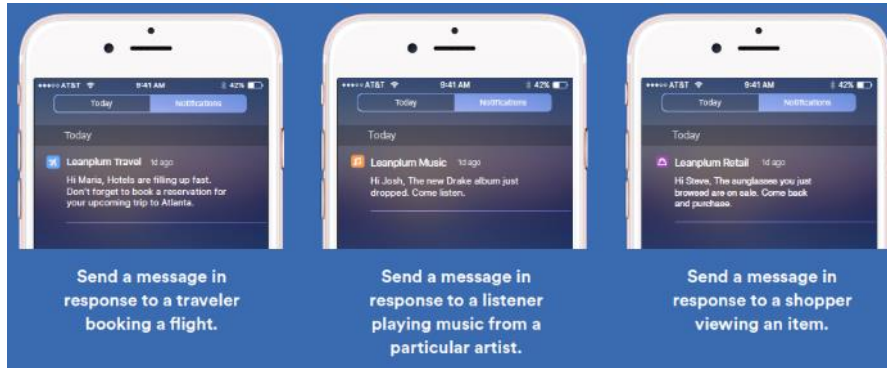
Gate game
rewards behind
value-added activity

Mobile Targeting: Personalization on Location/Situation

Mobile Targeting: Location & Situation Based Targeting

Mobile push notifications are created by specific situations and instant (real-time) actions of customers:

- Location / Whereabouts / Situation (i.e. jammed traffic)
- Recent Behavior: clicks, views, purchases
- Time of the day: (i.e. back home from work)
- Weather: rainy, snowy, sunny, warm or cold weather

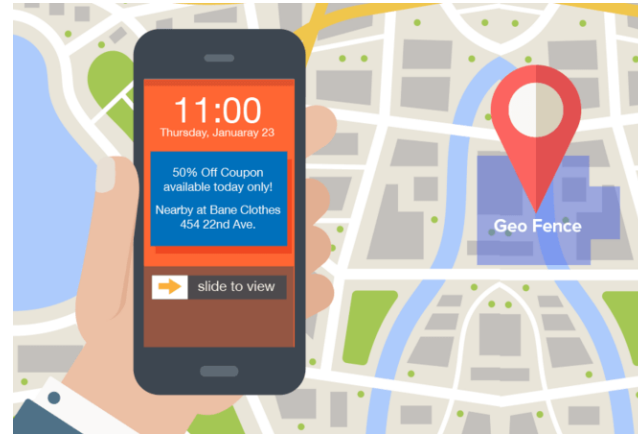


Mobile Targeting: Location & Situation Based Targeting

Mobile Targeting or Geomarketing (targeting) is one of the fastest growing domains in digital marketing.

The location of the potential customers are taken – together with other factors such as weather, previous clicks as targeting base

There are specific forms such as **ZIP-Code Targeting** and **Radius Targeting**



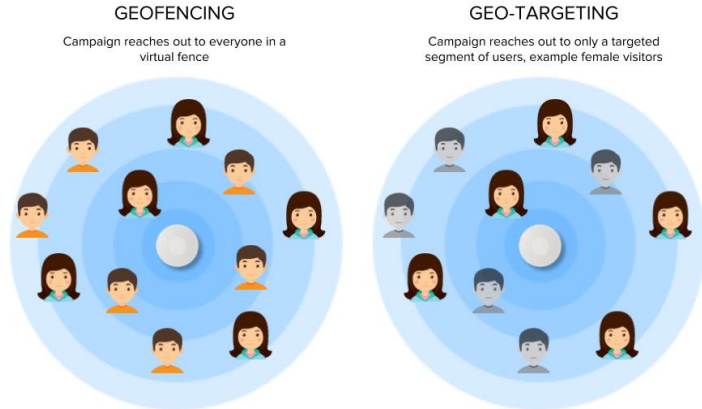
Mobile Targeting: Location Based (Geo) Targeting

Geofencing: sending mobile ads to **everyone** in a virtual fence at a certain time.

Geotargeting: send ads only to **potential (certain)** customers who are currently inside the “fence” or have been inside recently.

Geoconquesting reaching customers who are considering your competitors (nearby competitors’ locations) and targeting them

Beaconing: involves the use of beacons - a small, physical object that receives location data from nearby devices (permission based and precise)



Beacons

- Narrowest of three location-based marketing techniques
- Small bluetooth receiver that triggers messages in nearby devices
- Great for knowing if someone is near a specific object / product

LEANPLUM



Mobile Targeting: Location Based (Geo) Targeting

If you want to reach everyone in a certain area >

If you want to certain people in a certain area (i.e. females) >

If you want to reach/engage with your competitors' customer base >

If you want to interact with your customers in your locations >

Geofencing

Geotargeting

Geoconquesting

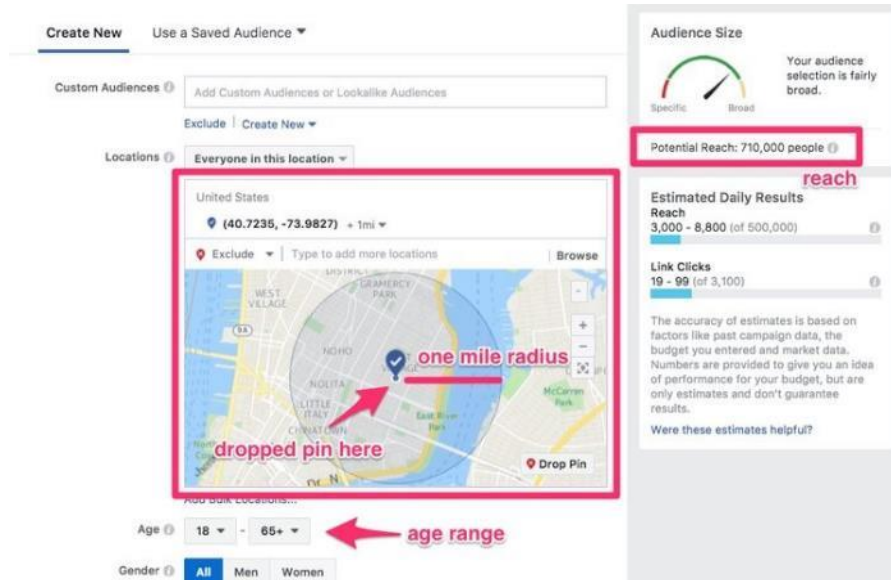
Beaconing

Geotargeting	Geofencing	Beacons
Best use for targeting people in a defined area	Best use for targeting people coming in, going out or dwelling inside a virtual fence	Best use for targeting nearby people who you would like to interact with
Generally used for a city or a specific location	Can be used anywhere where your users go, even around your competitors	Preferred in crowded or moving vehicles or in large retail stores that have aisles
Works on the basis of user device's last known location	Relies on the GPS of the device	Works only if the device Bluetooth is turned on
For example, all the 32-year-old males who are only in Manhattan and showed interest in a particular item can be targeted	For example, a retail store can run a flash sale and anybody who is nearby can get a notification	For example, in a trade fair attract people to your stall by sending exciting offers who are just standing outside

Facebook Geotargeting Advertising - An Example

Facebook Ads-Manager is used to reach customers/potential customers on the basis of their whereabouts through Facebook.

Multiple ways to refine audience by location, such as people living in or recently in a location, or people who are travelling in the location.



The screenshot shows the Facebook Ads Manager interface for creating a new audience. The 'Locations' section is highlighted with a red box, showing a map of the United States with a pin dropped in the Chicago area. A red circle indicates a 'one mile radius' around the pin. The 'Audience Size' section shows a 'Potential Reach' of 710,000 people. The 'Estimated Daily Results' section shows 'Reach' of 3,000 - 8,800 (of 500,000) and 'Link Clicks' of 19 - 99 (of 3,100). The 'Age' range is set to 18 - 65+, and the 'Gender' is set to All. A red arrow points to the 'age range' text.

Create New Use a Saved Audience ▼

Custom Audiences ⓘ Add Custom Audiences or Lookalike Audiences

Exclude Create New ▼

Locations ⓘ Everyone in this location ▼

United States
📍 (40.7235, -73.9827) + 1mi ▼

Exclude ▼ Type to add more locations Browse

Map showing a pin dropped in the Chicago area with a one mile radius circle around it. Text: "dropped pin here" and "one mile radius".

Age ⓘ 18 - 65+ ▼

Gender ⓘ All Men Women

Audience Size

Your audience selection is fairly broad.

Potential Reach: 710,000 people ⓘ

Estimated Daily Results

Reach
3,000 - 8,800 (of 500,000) ⓘ

Link Clicks
19 - 99 (of 3,100) ⓘ

The accuracy of estimates is based on factors like past campaign data, the budget you entered and market data. Numbers are provided to give you an idea of performance for your budget, but are only estimates and don't guarantee results.

Were these estimates helpful?

Enter the geographic information such as the address or target location

Specify the information for who you want to target, such as gender, age, languages, demographics, interest or behaviours.

Your Facebook geotargeting ads are published according to these settings.



(Responsible) Digital Marketing?

Towards Responsible (Digital) Marketing

Responsible marketing refers to the adoption of strategies that create **responsible outcomes**:

Ensuring that these practices **do not just pursue profitability but also cater to the well-being of society at large**.

Prioritizing **good citizenship, stakeholder orientation, and proactive responsibility** beyond mere compliance with the law.

Good for the world			
Good for the firm		No	Yes
	No	Lose-Lose How to avoid myopic marketing actions?	Lose-Win How to sustain marketing actions that are financially unviable, but can do good?
	Yes	Win-Lose How to expose and avoid the dark side of marketing?	Win-Win How to enable marketing actions that benefit multiple stakeholders?

Figure 1. The impact of marketing.

Note: "Good" is defined as long-term positive outcomes.



Source: Saren, M., Hassan, L. M., McGowan, M., Smith, N. C., Surman, E., & Varman, R. (2024). Introduction: Responsible marketing. In *Responsible Marketing for Well-being and Society* (pp. 1-22). Routledge.

Chandy, R. K., Johar, G. V., Moorman, C., & Roberts, J. H. (2021). Better marketing for a better world. *Journal of Marketing*, 85(3), 1-9.

Thank You